

A cool paper about graph rag

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Abstract

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Keywords: retrieval-augmented generation, graph RAG, incremental ingestion, evaluation, knowledge graphs

1. Introduction

Retrieval-Augmented Generation (RAG) has emerged as a foundational technique for grounding large language models in external knowledge (...) Standard RAG approaches (describe: retrieval of fixed-length text chunks via vector similarity. state limitations)

Graph-based RAG methods were introduced to (address limitations).

GraphRAG methods have evolved rapidly, with contemporary methods ... (introduce modern methods with their unique contributions) This shift fundamentally changes what it means to "retrieve context". In modern GraphRAG systems, retrieved context may consist of hierarchical summaries, merged relation descriptions, hypergraph constructions, and subgraph projections rather than contiguous spans of source text.

As these methods have grown more capable, evaluation frameworks (.. are getting left behind- measuring accuracy only is not effective). A system that returns the entire source corpus as context will answer nearly every evaluation question correctly, but ...

...

In this paper, we conduct a systematic comparison of three advanced GraphRAG systems from recent literature: KGen, HyperGraphRAG, and LeanRAG. We compare these methods alongside two baseline techniques- a standard vector-based RAG system and a method we call "Parrot", which returns the full source text for every query.

Our investigation of standard evaluation methods show that existing benchmarks are insufficiently challenging on the widely-used MINE dataset ...

... To address the inadequacy of accuracy as a sole metric, we introduce conciseness ...

2. RAG Methods

Here we discuss each method as encoding + retrieval + context synthesis ... (discuss the commonalities) - corpus -> chunking -> embeddings -> method specific structure of building the knowledge graph - query -> method specific retrieval -> retrieved context string -> evaluation techniques

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Parrot - Def: returns full article text as retrieved context - Matters because it distinguishes "model can answer if information is present" from "retrieval found it" - Expect to see that parrot saturates inclusion but with a massive mean context length and therefore poor conciseness.

VectorRAG - Standard chunk retrieval mechanism - Define what VectorRAG settings we have to parameterize with

KGGen - Describe KGGen's multi stage extraction, aggregation, entity + edge resolution, motivation for avoiding conflation and reducing sparsity - Define our kg gen implementation - Note that poor inclusion may indicate representational loss in triples for complex facts

HyperGraphRAG - Describe the shift in representation: hyperedge construction, connecting entities and describing in natural language - Define our hgr implementation

LeanRAG - Describe LeanRAG's introduction of hierarchical summarization, inter cluster relations, .. - Bottom up LCA retrieval mechanism - The core property of leanrag is that it's designed to reduce redundancy but maintain quality

3. Experiments

Discuss our datasets here that we used: - MINE: ... - QASPER: ...

Also discuss the metrics that we use here plus general setup for our experiments

4. Results

Name	Score	Context Length (chars)	Conciseness
parrot	99.37% (790/795)	63409.93	0.00156712
vector_200_20_95	13.46% (107/795)	3859.23	0.00348751
vector_200_20_85	34.09% (271/795)	11511.37	0.00296125
vector_200_0_95	11.32% (90/795)	3601.37	0.00314346
vector_100_20_95	12.96% (103/795)	4297.10	0.00301505
vector_100_20_85	throws	an	error!
vector_100_0_95	12.20% (97/795)	4077.16	0.00299258
hgr_60_50_60_5	30.69% (244/795)	692.71	0.04430669
hgr_60_30_60_3	45.28% (360/795)	1435.07	0.03155464
hgr_dump	52.08% (414/795)	6044.61	0.00861520
kg-gen	12.58% (100/795)	909.92	0.01382387
kg-gen_dump	14.21% (113/795)	3979.79	0.00357151
leanrag-ours-1	97.33% (511/525)	31375.23	0.00310223

Table 1. MINER evaluation results

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5. Conclusion